

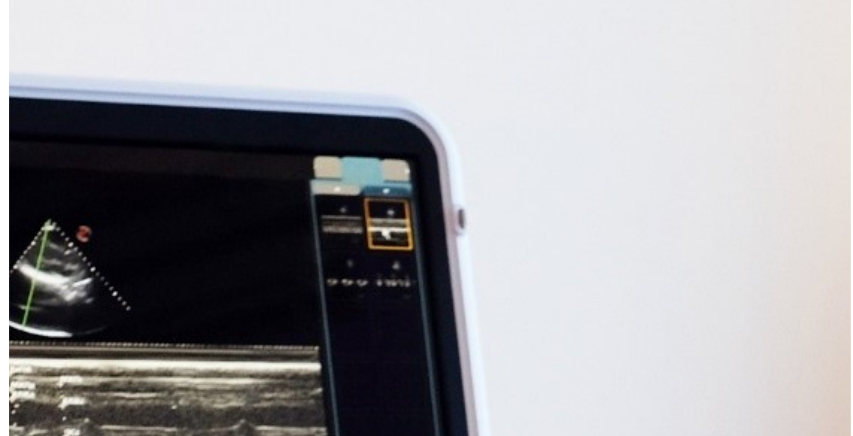


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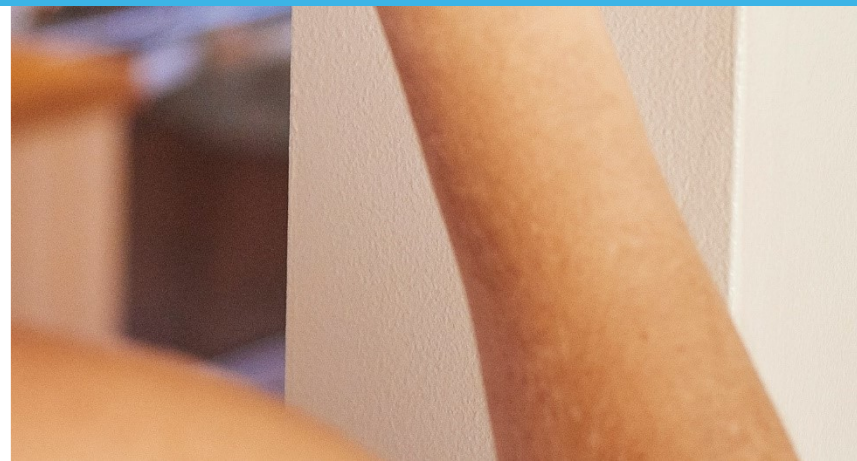


STM32H7

MCUs for rich and complex applications



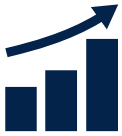
STM32H7





STM32H7 series

New product lines expanding the STM32 portfolio



New Performance Record

Up to 2424 + 800 CoreMark (Cortex[®]-M7 @480MHz + Cortex[®]-M4 @240MHz) in Dual core
Up to 2778 CoreMark (Cortex[®]-M7 @550MHz) in Single core



Single and Dual-core flexible architecture for industrial, security or AI applications
Accelerated graphics, fast data transfer, advanced peripherals



Advanced security features

Crypto Hash, Cortex[®]-M7 Security services

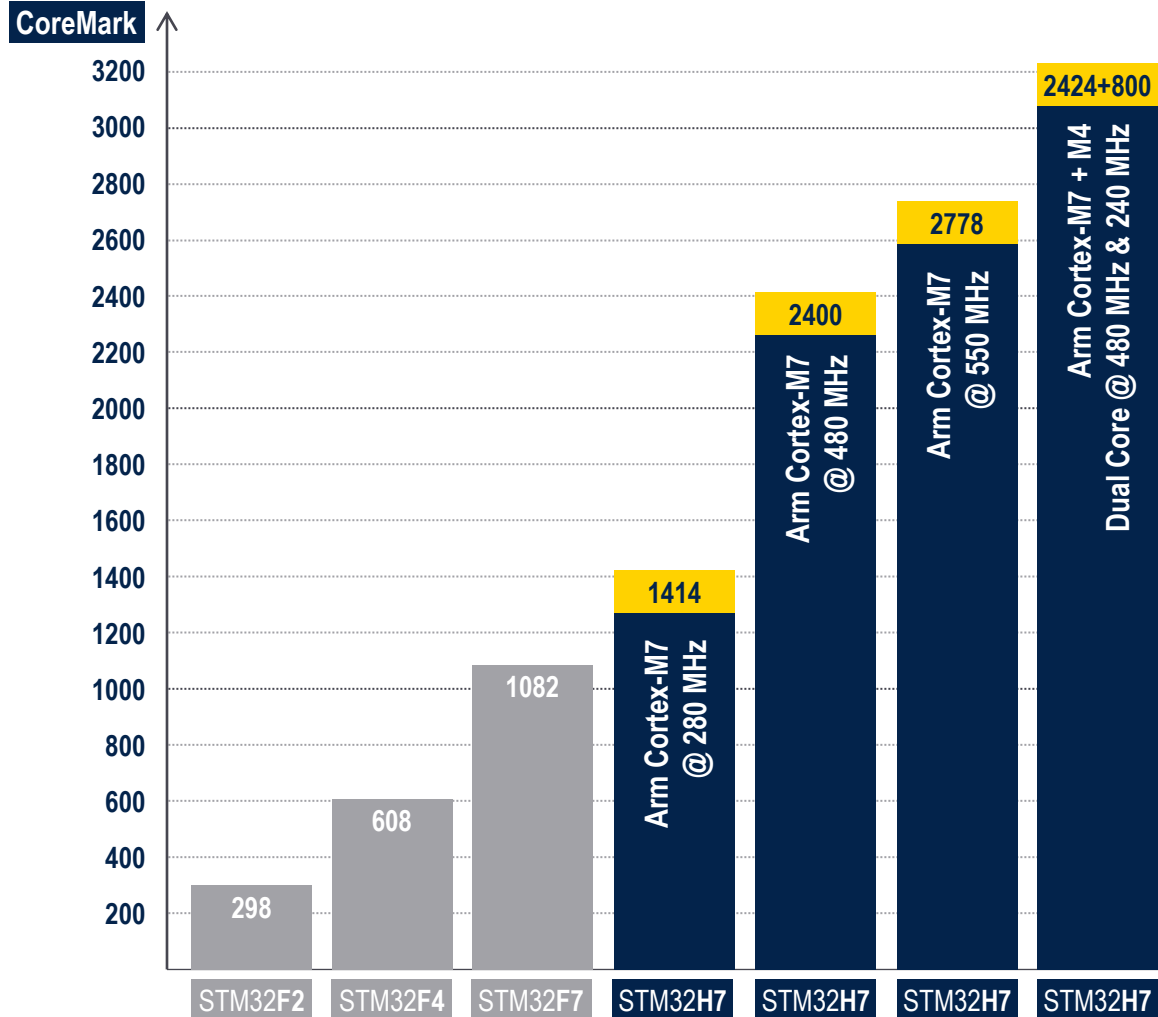


Rich eco-system to speed-up your design

SW tools, HW boards, community and partners

Performance record





High performance range

Arm® Cortex® -M7 up to 550 MHz

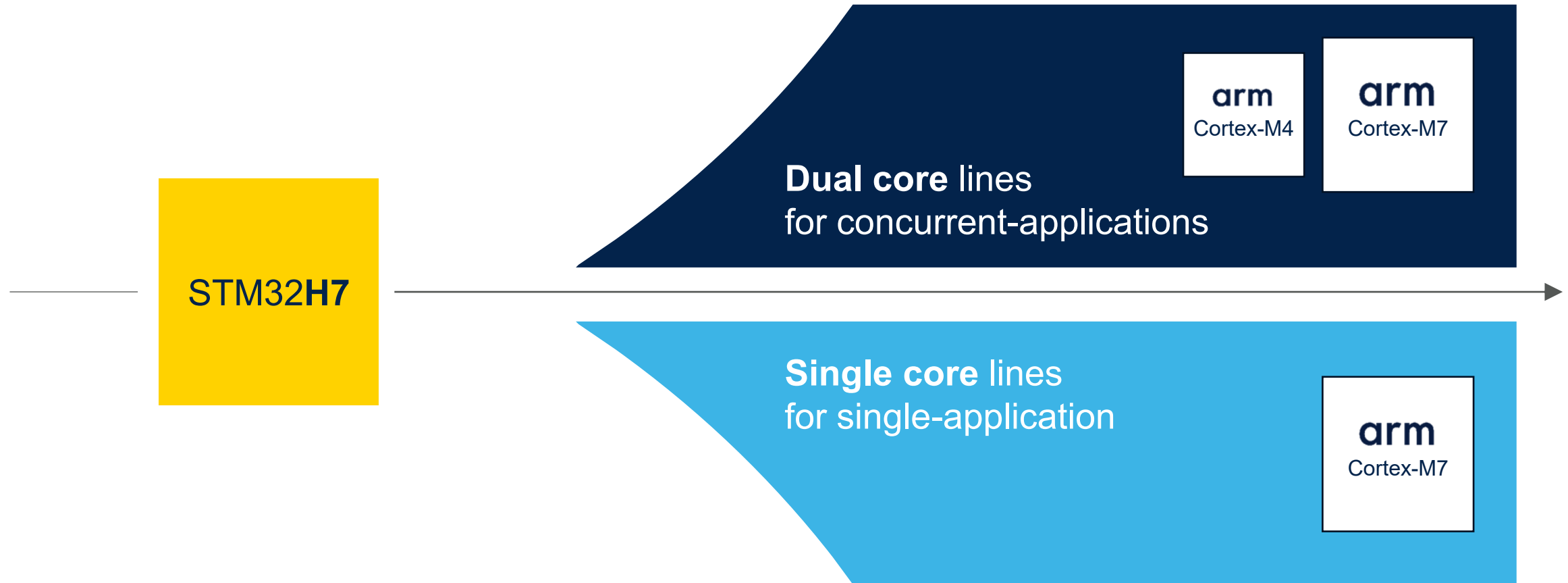
Most powerful Cortex core with double precision FPU, MPU, advanced DSP and L1 cache

Arm® Cortex® -M4 @240 MHz

Best in class core for **real-time** with single precision FPU, DSP, MPU and ART Accelerator™



The extended STM32H7 experience



Single Core Architecture Approach for performance and advanced HMI

Factory automation

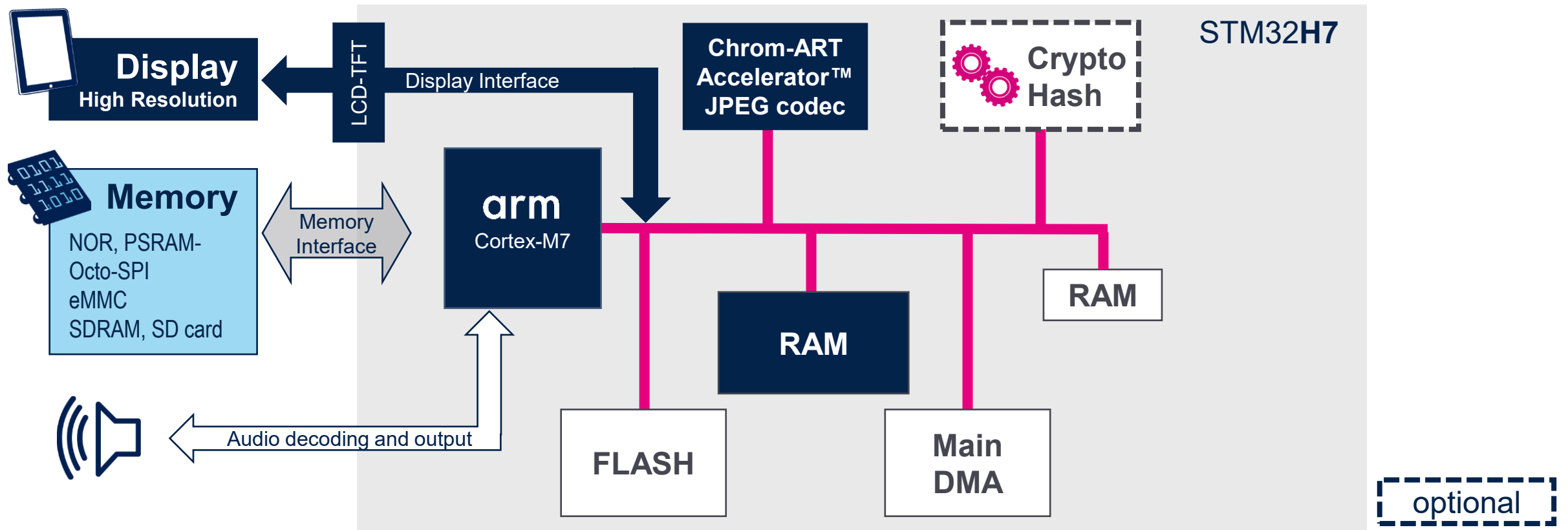


Connectivity & security



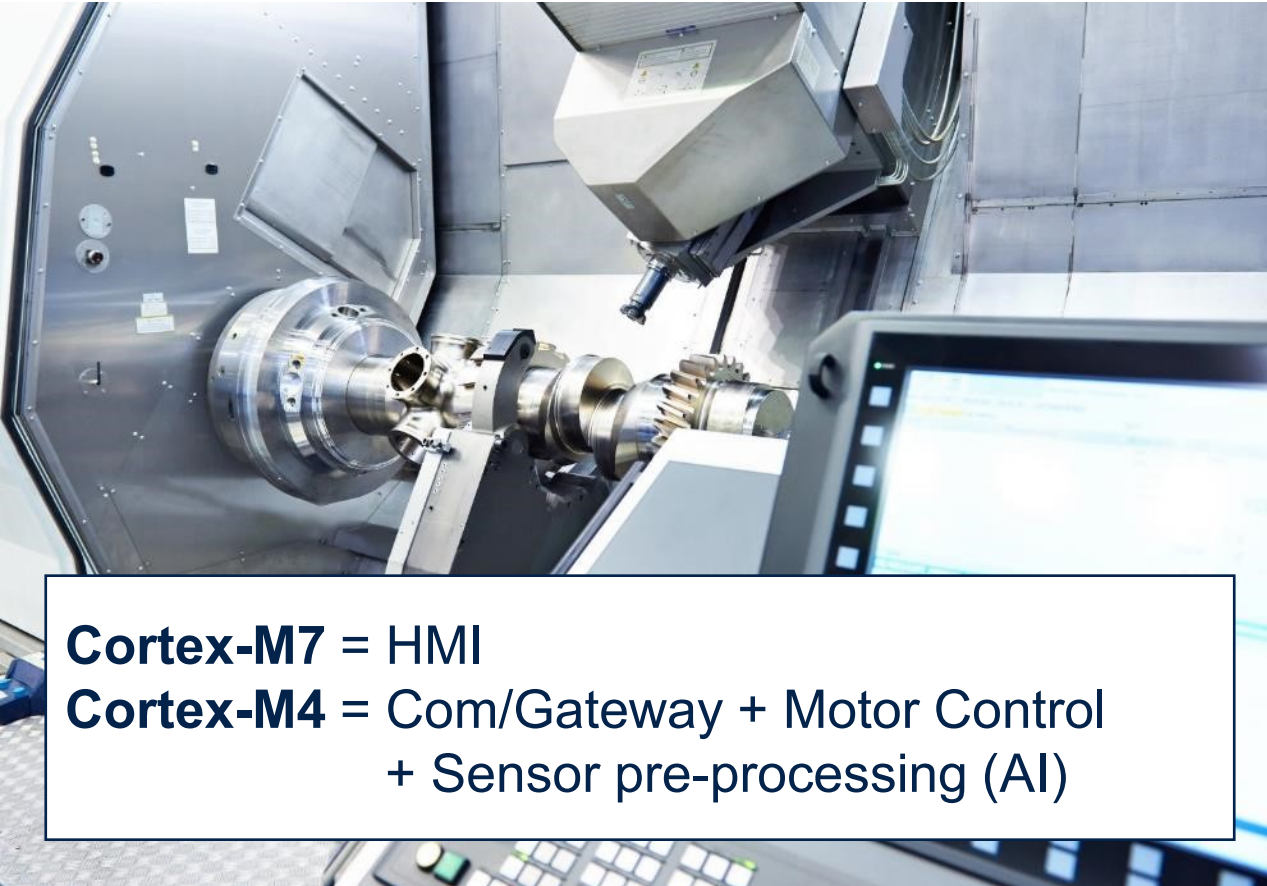
Create a rich human machine interface

Cortex-M7 - handling audio and rich HMI, Real Time control tasks



Dual-core architecture approach for richer and more complex applications

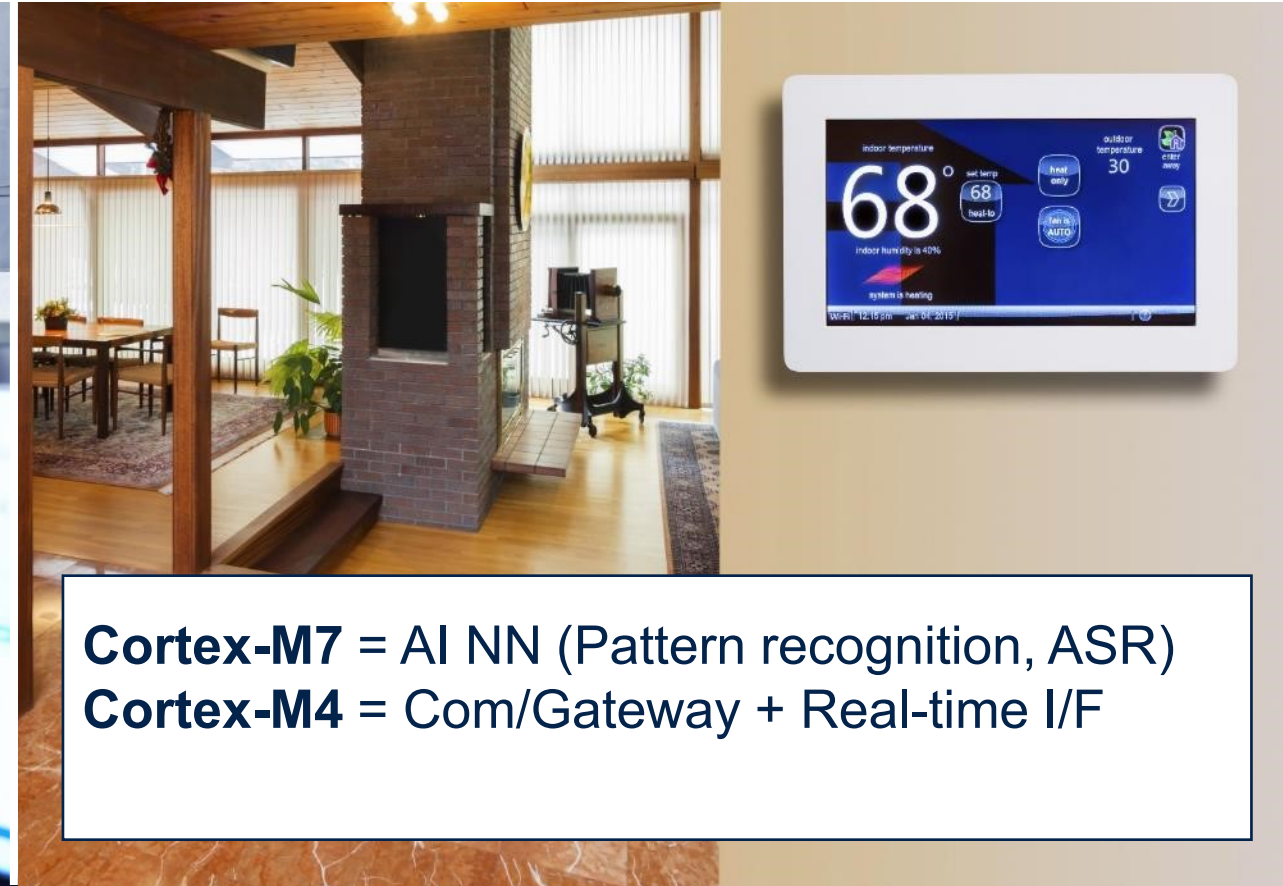
Industrial tool machine



Cortex-M7 = HMI

Cortex-M4 = Com/Gateway + Motor Control
+ Sensor pre-processing (AI)

Home automation & security



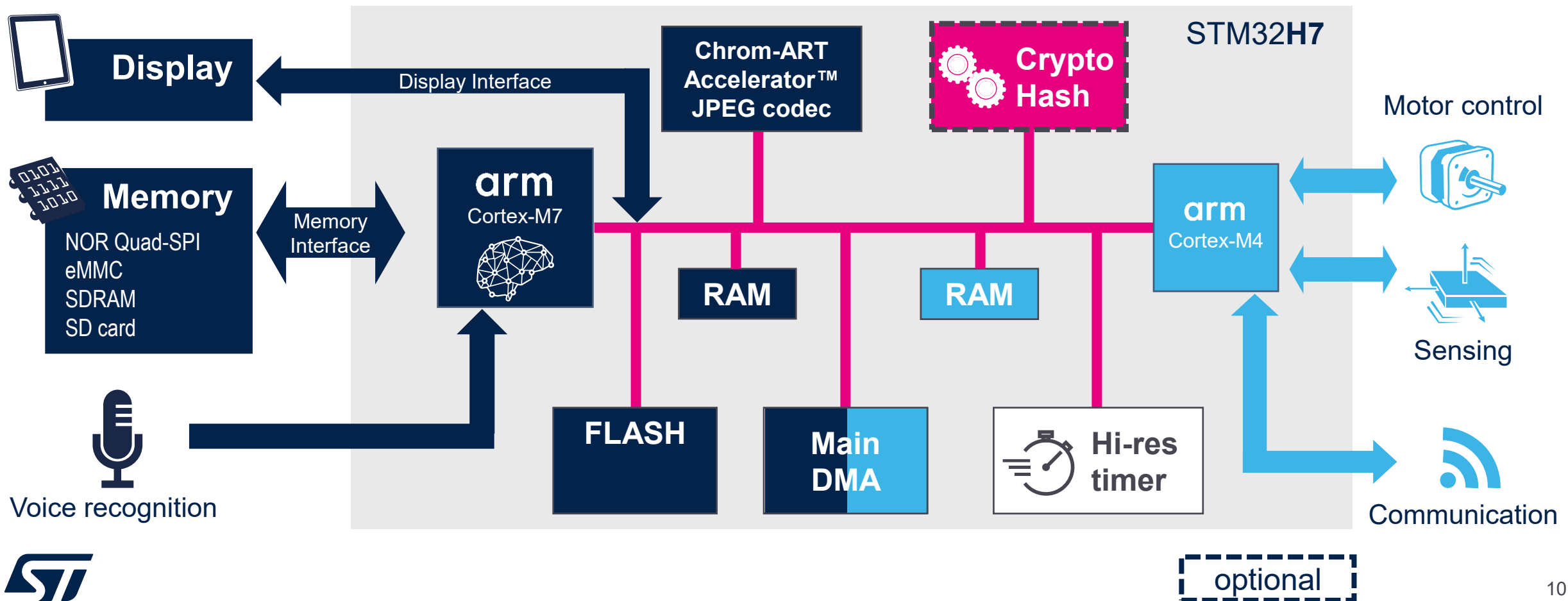
Cortex-M7 = AI NN (Pattern recognition, ASR)

Cortex-M4 = Com/Gateway + Real-time I/F



Build complex applications mixing AI and real-time control

Connected Kitchen Aid with advanced HMI (Large display and Voice recognition)





Powerful cores supported by a powerful architecture

Display nice graphic

The Chrom-ART Accelerator™
and MJPEG codec
offload the CPU by more than 90%



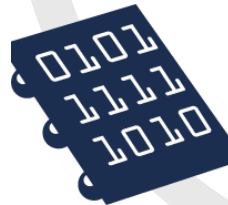
Manage security

Use dedicated **cryptography**
and **Hashing** HW acceleration
to offload the CPU by more than 90%



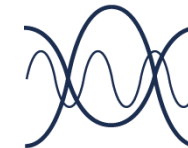
Transfer data efficiently across peripherals

The Main DMA takes care
of the most complex schemes
between memories and peripherals
with **up to 16 channels to offload the CPU**



Generate complex wave forms

High-Resolution timer (2.1ns)
can generate complex wave forms
synchronized on multiples events,
with no CPU assist



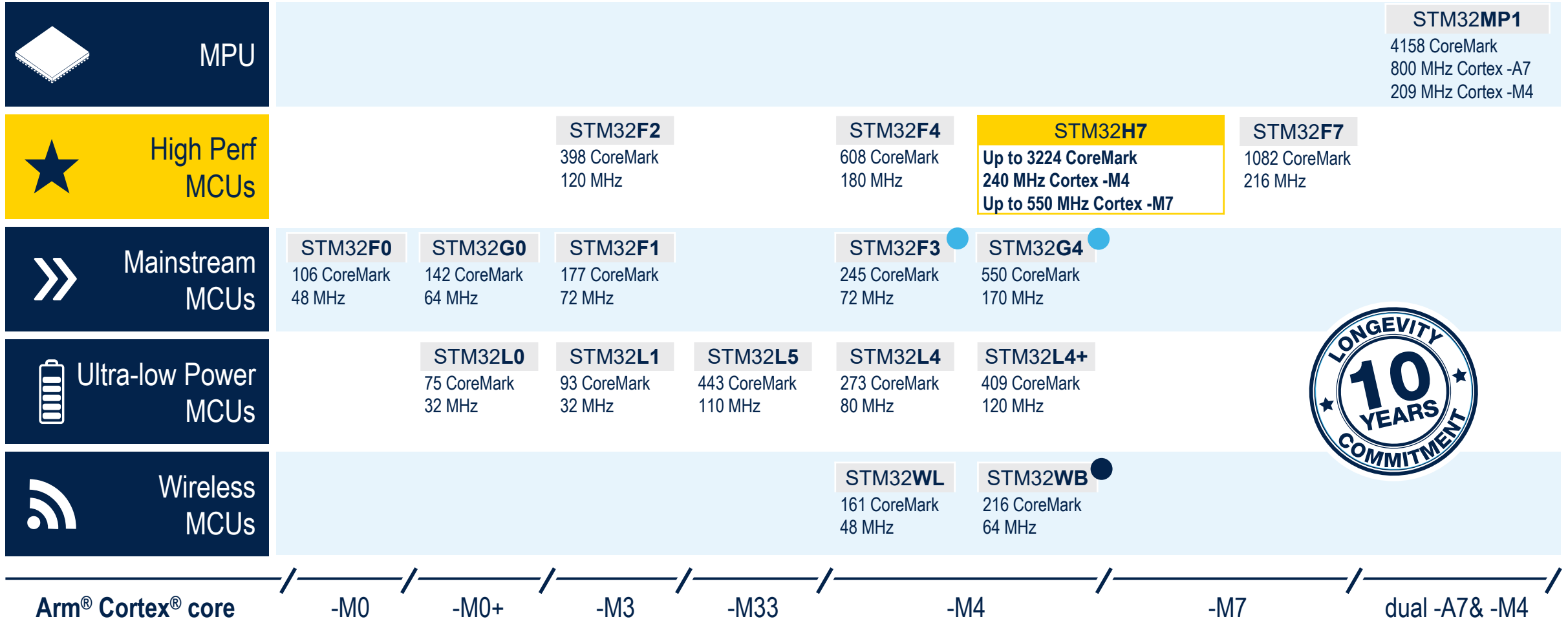
STM32H7

STM32H7 line-up





STM32H7 MCU series



● Optimized for mixed-signal applications

● Cortex-M0+ Radio co-processor



Extensive STM32H7 portfolio



Dual-core Line

STM32H745/755
480+240 MHz **SMPS**
1027 + 300 DMIPS
RAM 1 MB
Flash up to 2 MB

STM32H747/757
480+240 MHz **SMPS**
1027 + 300 DMIPS
RAM 1 MB
Flash up to 2 MB



Single-core Line

STM32H7A3/B3
280 MHz **LDO**
599 DMIPS
RAM 1.4 MB
Flash up to 2 MB

STM32H742
480 MHz **LDO**
1027 DMIPS
RAM 692 KB
Flash up to 2 MB

STM32H743/753
480 MHz **LDO**
1027 DMIPS
RAM 1 MB
Flash up to 2 MB

STM32H723/733
550 MHz **LDO**
1177 DMIPS
RAM 564 KB
Flash up to 1 MB

STM32H725/735
550 MHz **SMPS**
1177 DMIPS
RAM 564 KB
Flash up to 1 MB



Extended temperature
range 125 °C ambient



Value Line

STM32H7B0
280 MHz **LDO**
599 DMIPS
RAM 1.4 MB
Flash 128 KB

STM32H750
480 MHz **LDO**
1027 DMIPS
RAM 1 MB
Flash 128 KB

STM32H730
550 MHz **LDO**
1177 DMIPS
RAM 564 KB
Flash 128 KB

STM32H730Q
550 MHz **SMPS**
1177 DMIPS
RAM 564 KB
Flash 128 KB

Arm® Cortex® core

Cortex-M7

Cortex-M7 & -M4



STM32H7 MCU Series 32-bit Arm® Cortex®-M7 or Cortex®-M7 + Cortex®-M4



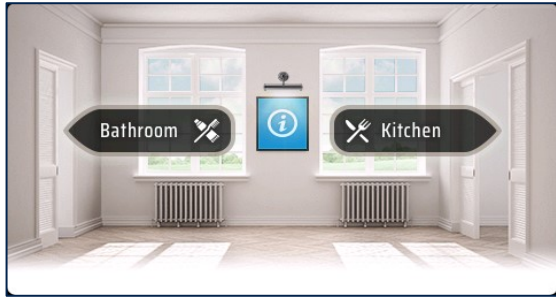
Tailored for your needs

CORE, MEMORIES AND ACCELERATION	Product line	f _{CPU} (MHz)	Dual- Bank Flash memory (bytes)	RAM (bytes)	OctoSPI & OTFDEC ³	Ethernet	Graphic	Power supply	Stop mode (typical) / RAM reten- tion
- Single-core Cortex-M7 up to 550 MHz - Dual-core Cortex-M7 480 MHz and Cortex-M4 240 MHz - Flash and RAM acceleration - SP-FPU and DP-FPU 4 x DMA - Mathematics (only H723/733/725/735/730)	Dual-core lines								
	STM32H747/757 ¹	480 + 240	Up to 2 Mbytes	1 Mbyte (incl.128 Kbytes DTCM + 64 Kbytes ITCM + 64 Kbytes backup ¹) + 4 Kbytes backup ²		•	TFT-LCD JPEG codec MIPI-DSI	SMPS + LDO	360 µA / 1MB 250 µA / 768KB
CONNECTIVITY - Up to 2 x USB2.0 OTG FS/HS 2 x SDMMC - USART, UART, SPI, I2C - Up to 3 x CAN (2 x FD and 1 x TT) - HDMI-CEC - FMC, Dual-mode Quad-SPI or 2 x Octo-SPI - Camera I/F	STM32H745/755 ¹	480 + 240	Up to 2 Mbytes	1 Mbyte (incl.128 Kbytes DTCM + 64 Kbytes ITCM + 64 Kbytes backup ¹) + 4 Kbytes backup ²		•	TFT-LCD JPEG codec	SMPS + LDO	360 µA / 1MB 250 µA / 768KB
	Single-core lines								
AUDIO 3 x PS + audio PLL 4 x SAI 2 x 12-bit DAC - SPDIF-RX	STM32H7A3/7B3 ¹	280	Up to 2 Mbytes	1.4MB (incl.128K DTCM, 64K ITCM, 1184K+SRAM, 4K backup)	•		TFT-LCD JPEG codec Chrom- GRC	SMPS + LDO	32 µA / 1.4MB 28 µA / 32KB
	STM32H743/753 ¹	480	Up to 2 Mbytes	1 Mbyte (incl.128 Kbytes DTCM + 64 Kbytes ITCM + 64 Kbytes backup ¹) + 4 Kbytes backup ²		•	TFT-LCD JPEG codec	LDO	1270 µA / 1MB 910 µA / 768KB
GRAPHIC Chrom-ART Accelerator™	STM32H742	480	Up to 2 Mbytes	692 Kbytes (incl.128 Kbytes DTCM + 64 Kbytes ITCM + 16 Kbytes backup ¹) + 4 Kbytes backup ²		•		LDO	1270 µA / 692KB 910 µA / 704KB
	STM32H725/735 ³	550	Up to 1 Mbyte	564KB (incl.128K DTCM, 432KB Syst + 4K bckup)	•	•	TFT-LCD	SMPS ⁵ + LDO	200 µA / 564KB
OTHER - Crypto/Hash option (except H742) - Security services option (except H742) - TRNG - DFSDM 16- and 32-bit timers - HRTimer (except STM32H7A7/H7B7/H7B0/H723/H725/H730/H733/H735) - Up to 3 x 16-bit ADC (up to 3.6 MSPS) - Analog (compt.AOP) - Voltage range 1.62 to 3.6 V (except 100-pin and VFQFN68 packages : 1.71 to 3.6 V) - Multi-power domains -40°C up to 105°C ambient -40°C up to 125°C ambient²	STM32H723/733 ³	550	Up to 1 Mbyte	564KB (incl.128K DTCM, 432KB Syst + 4K bckup)	•	•	TFT-LCD	LDO	520 µA / 564KB
	Value line								
	STM32H7B0	280	128 Kbytes	1.4MB (incl.128K DTCM, 64K ITCM, 1184K+SRAM, 4K backup)	•		TFT-LCD JPEG codec Chrom- GRC	SMPS + LDO	32 µA / 1.4MB 28 µA / 32KB
	STM32H750	480	128 Kbytes	1 Mbyte (incl.128 Kbytes DTCM + 64 Kbytes ITCM + 64 Kbytes backup ¹) + 4 Kbytes backup ²		•	TFT-LCD JPEG codec	LDO	1270 µA / 1MB 910 µA / 768KB
	STM32H730	550	128 Kbytes	564KB (incl.128K DTCM, 432KB Syst + 4K bckup)	•	•	TFT-LCD	SMPS ⁴ + LDO	200 µA / 564KB 520 µA / 564KB

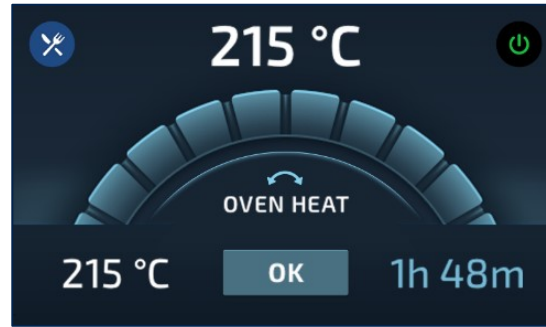
- Single and Dual core versions
- High performance up to 480 MHz in Dual core and up to 550 MHz in Single core
- 128 KB to 2 MB Flash Dual Bank
- Up to 1.4 MB RAM
- More security features (Boot, Tamper ...), OTFDEC on external memories, Crypto/Hash and security services (optional)
- 35 communication peripherals
- 16-bit ADC up to 3.6 Msps, up to 5 MSPS in 12-bit, Comparators, Op Amp
- TT-FD-CAN and FD-CAN
- High-Resolution timer (2.1ns)
- Low-Power Timers
- LDO and SMPS option
- Up to 140 °C junction temperature / 125 °C ambient (optional)

Notes :
1. Optional - dedicated CPM, STM32H733, STM32H735, STM32H753, STM32H755, STM32H757, STM32H7B3 for the Crypto Variants
2. 125 °C ambient / 140 °C junction. Dedicated part numbers on STM32H725/H735, STM32H745/H755
3. Crypto and Security services on CPM : STM32H733, STM32H735 and STM32H730
4. SMPS available only on STM32H730Q CPM
5. SMPS only on the QFN68 variant (no LDO)

Enhance your product with great graphic



 Smart home
[Watch video](#)



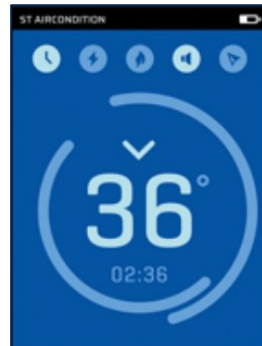
 Kitchen appliances
[Watch video](#)



 Industrial
[Watch video](#)



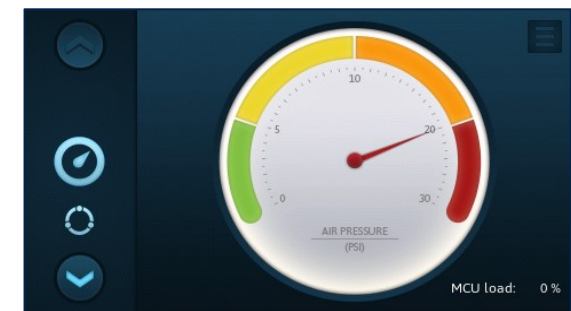
 Smart home
[Watch video](#)



 Smart home
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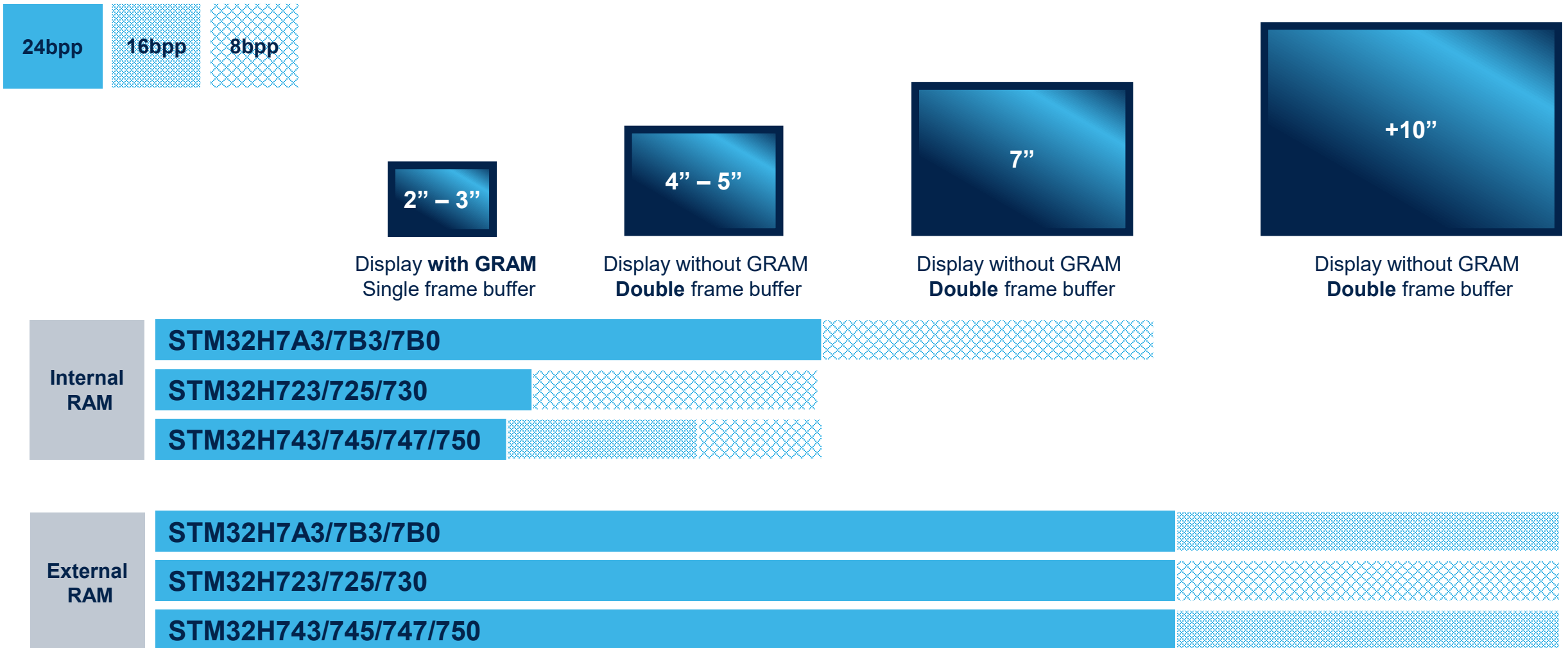


 Industrial
[Watch video](#)



 Industrial
[Watch video](#)

Enhanced graphic UI for any resolution



X-CUBE-TouchGFX

Graphical User Interface development

Faster and easier GUI Development

Free for all STM32 Developers

Maximum Performance on minimum footprint

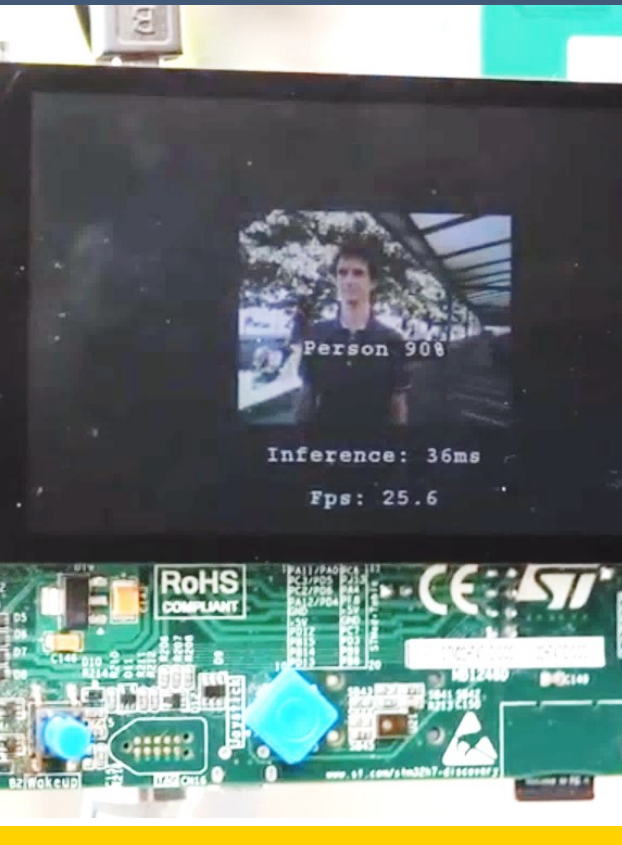
Interoperable with STM32Cube Ecosystem



TouchGFX

Embed AI in your applications with STM32H7

People detection



Food classification



Voice recognition

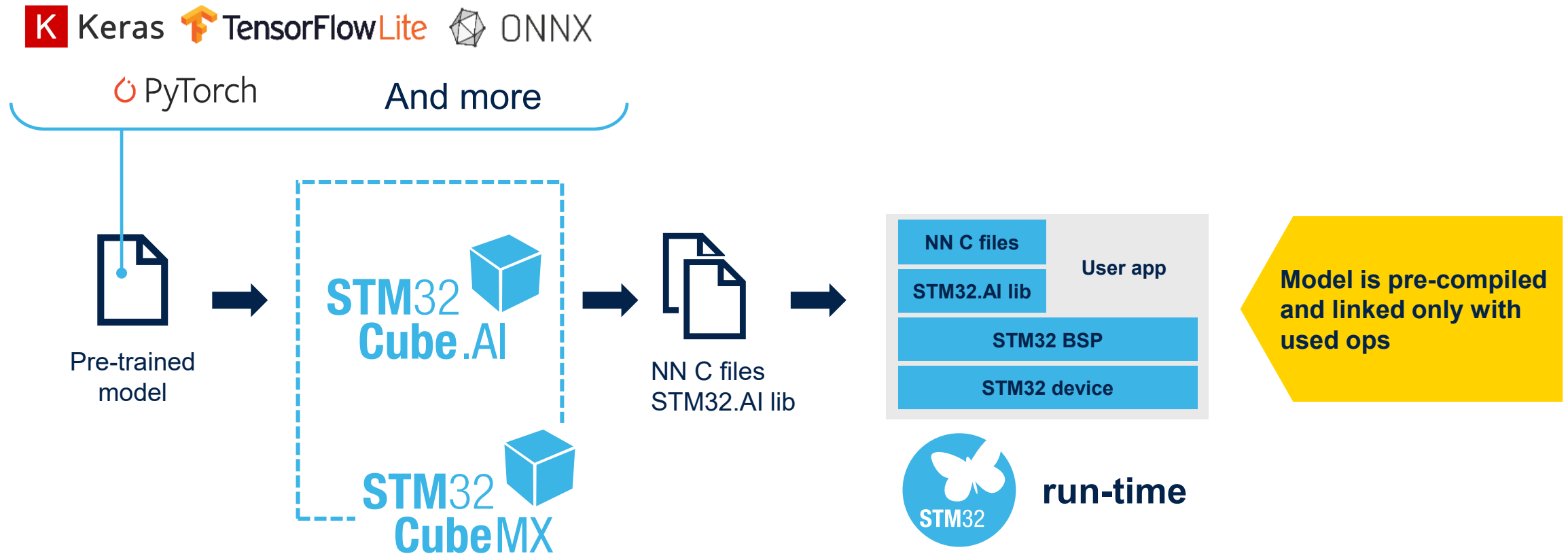


Industrial applications



The key steps behind Neural Networks on STM32

Optimized C code generated by STM32Cube.AI





SIL functional safety package for STM32

Reduce time and cost to build STM32-based systems certified to IEC 61508 industrial safety standard





STM32 high performance built-in safety features

Features	STM32F2/F4	STM32F7	STM32H7
Dual watchdogs: Independent watchdog and system window watchdog	•	•	•
Backup clock circuitry with clock security system (CSS)	•	•	•
Hardware CRC unit / Programmable polynomial	• / -	• / •	• / •
Supply monitoring (POR, BOR, PVD)	•	•	•
I/O function locking	•	•	•
PWM critical register protections (write-once registers)	•	•	•
Memory protection unit (MPU) 8 zones – to ensure data integrity from invalid behavior	•	•	•
Multiple Flash memory protection levels	•	•	•
ECC Error Code Correction (SECEDED) for SRAM	-	-	•
ECC Error Code Correction (SECEDED) for Flash memory	-	-	•

Note: Arm Cortex-M cores also have built-in safety features (dual stack pointer, fault exceptions, and debug module).



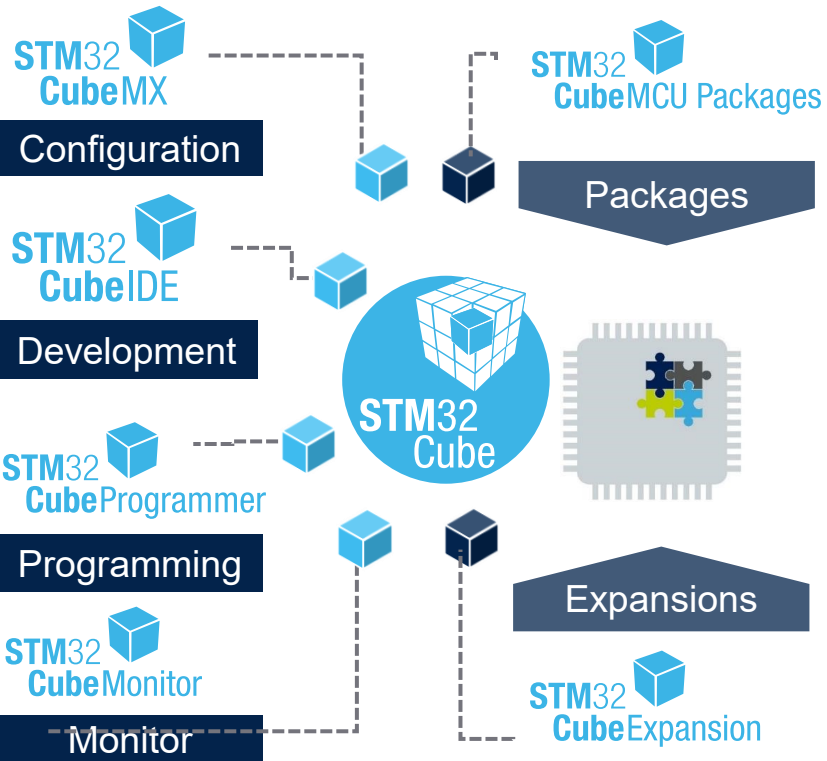
Supported by the STM32 ecosystem

Software



Software
Tools

Embedded
Software



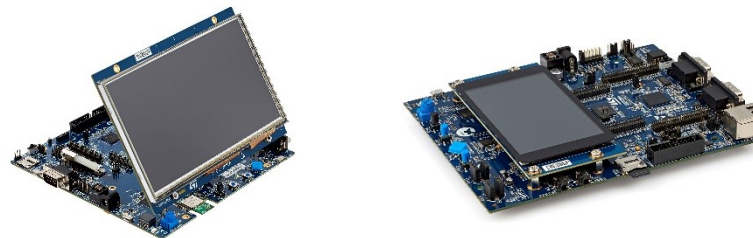
Hardware



STM32 Nucleo boards



Discovery kits



Evaluation boards

Customer support



FAE - Worldwide
Customer Support



community.st.com

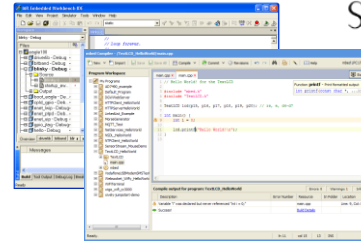
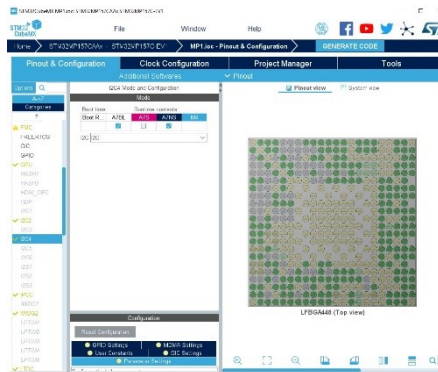
**MO
OC**





Software tools for STM32H7

Complete support of Arm Cortex-M architecture



STM32CubeMX

STM32CubeMX enhanced for Dual-core

- Configure and generate Code
- Multi-core resources allocation
- Peripherals configuration

IDEs Compile and Debug

Multi-Core Solutions

- Partners IDE
- Free IDE based on Eclipse
- **Multi-core** debugging

STM32 Programming & Monitoring tools

STM32CubeProg STM32CubeMonitor

- Program the application into the chip
- Sign the application and generate license
- Monitor variables at run-time



STM32H7 hardware solutions

Speed-up evaluation, prototyping and design
(board selection guide available at the end of this presentation)



5 Evaluation Boards

Full feature STM32H7 evaluation

6 Discovery Kits

Flexible prototyping & demo

6 Nucleo Boards

Affordable and quick prototyping



Pick the right STM32H7 development tool

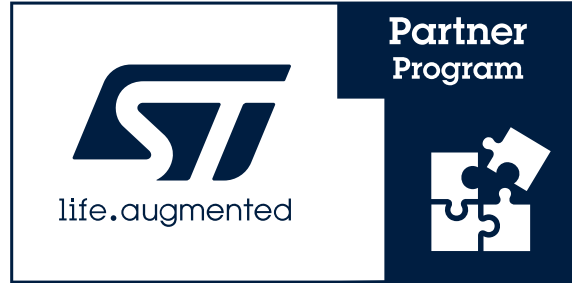


STM32H7 class	Cores/Speed	Part numbers	Evaluation boards	Discovery Kits	Nucleo boards
STM32H74/5	Single Core 480 MHz	STM32H743	STM32H743I-EVAL2	-	NUCLEO-H743ZI2
		STM32H753, Crypto enabled	STM32H753I-EVAL2	-	NUCLEO-H753ZI
		STM32H750 Value Line, Crypto enabled	-	STM32H750B-DK	-
	Dual Core 480 MHz + 240 MHz	STM32H745	-	STM32H745I-DISCO	NUCLEO-H745ZI-Q
		STM32H747	STM32H747I-EVAL	STM32H747I-DISCO STM32H747I-DISC1	-
		STM32H755/757, Crypto enabled	STM32H757I-EVAL	-	NUCLEO-H755ZI-Q
STM32H7A/B	Single Core 280 MHz	STM32H7A3	-	-	NUCLEO-H7A3ZI-Q
		STM32H7B3, Crypto enabled	STM32H7B3I-EVAL	STM32H7B3I-DK	-
		STM32H7B0, Value line, Crypto enabled	STM32H7B3I-EVAL *	STM32H7B3I-DK *	-
STM32H72/3	Single Core 550 MHz	STM32H723/733	-	-	NUCLEO-H723ZG
		STM32H725/735	-	STM32H735G-DK	-
		STM32H730, Value line, Crypto enabled	-	STM32H735G-DK *	-

* Recommended board (no dedicated board for this part number)



Software, tools and services a broad ecosystem to support development



Large selection of partners
already engaged for:

- Embedded software
- Software tools
- Graphics UI
- Security
- Training and services





Releasing your creativity



[/STM32](#)



[@ST_World](#)



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[wiki.st.com/stm32mcu](#)



[github.com/STMicroelectronics](#)



[STM32H7 online training](#)



[STM32H7 blog article](#)



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